

Unit 2 - Number Theory in Cryptography

Cryptography and Web Security (DI03016021) - GTU Diploma IT Sem 3

16% weightage | 8 hours

How to use these notes: each topic is written the way you can copy it in the exam. For a **3-mark** question write only the definition + 2-3 points. For **4 marks** add an example or table. For **7 marks** write everything, with diagram and table. Lines marked ★ show where the topic was already asked in a GTU paper (only 2 papers exist so far: Winter 2025 and Summer 2026).

Questions already asked from this unit (do these first)

- Asked in W2025 Q2(a) [3M: divisibility and division algorithm]
- Asked in S2026 Q2(b) [4M: modular properties with example] + W2025 Q2(a)OR [3M: prove $(a \times b) \bmod n$]
- Asked in BOTH papers: W2025 Q2(b)OR [4M: $\mathbb{Z}_n$, $\mathbb{Z}_n^*$, members of $\mathbb{Z}_7$ and $\mathbb{Z}_7^*$] + S2026 Q2(b)OR [4M: residue set $\mathbb{Z}_n$]
- Euclidean algorithm asked in BOTH papers (7M each): W2025 Q1(c) + S2026 Q2(c)OR
- Extended Euclidean algorithm asked in BOTH papers (7M each): W2025 Q1(c)OR + S2026 Q2(c)

2.1 Divisibility and Division Algorithm

★ Asked in W2025 Q2(a) [3M: divisibility and division algorithm]

- **Divisibility:** an integer **a** is divisible by **b** (written $b \mid a$) if $a = b \times k$ for some integer k , i.e. the remainder is zero. *Example: $12 = 3 \times 4$, so $3 \mid 12$.*

Properties: if $a \mid b$ and $b \mid c$ then $a \mid c$; if $a \mid b$ and $a \mid c$ then $a \mid (m \cdot b + n \cdot c)$; if $a \mid b$ and $b \mid a$ then $a = \pm b$.

- **Division algorithm:** for any integer **a** and positive integer **n**, there are **unique** integers q (quotient) and r (remainder) such that

$$a = q \times n + r \quad \text{where } 0 \leq r < n$$

Examples: $17 = 3 \times 5 + 2$ ($q = 3, r = 2$). $-7 = (-3) \times 3 + 2$ ($q = -3, r = 2$). The remainder is never negative.

2.2 Modulo Operator and its properties

★ Asked in S2026 Q2(b) [4M: modular properties with example] + W2025 Q2(a)OR [3M: prove $(a \times b) \bmod n$]

Modulo operator: $a \bmod n$ gives the **remainder** when a is divided by n . *Example: $17 \bmod 5 = 2$.* If $a \bmod n = b \bmod n$ we write $a \equiv b \pmod{n}$.

Three properties (use any numbers - here $a = 12$, $b = 11$, $n = 7$)

Property	Left side	Right side	Result
Addition: $(a + b) \bmod n = [(a \bmod n) + (b \bmod n)] \bmod n$	$23 \bmod 7 = 2$	$(5 + 4) \bmod 7 = 9 \bmod 7 = 2$	Equal ✓
Subtraction: $(a - b) \bmod n = [(a \bmod n) - (b \bmod n)] \bmod n$	$1 \bmod 7 = 1$	$(5 - 4) \bmod 7 = 1$	Equal ✓
Multiplication: $(a \times b) \bmod n = [(a \bmod n) \times (b \bmod n)] \bmod n$	$132 \bmod 7 = 6$	$(5 \times 4) \bmod 7 = 20 \bmod 7 = 6$	Equal ✓

Prove: $a = 12, b = 11, n = 7 \rightarrow (a \times b) \bmod n = [(a \bmod n) \times (b \bmod n)] \bmod n$

LHS: $(12 \times 11) \bmod 7 = 132 \bmod 7$. Since $132 = 18 \times 7 + 6$, LHS = **6**.

RHS: $12 \bmod 7 = 5$ and $11 \bmod 7 = 4$. So $[(5) \times (4)] \bmod 7 = 20 \bmod 7 = \mathbf{6}$ (because $20 = 2 \times 7 + 6$).

LHS = RHS = 6. **Hence proved.**

General reason: write $a = n \cdot q_1 + r_1$ and $b = n \cdot q_2 + r_2$. Then $a \times b = n(\dots) + r_1 \times r_2$, so $a \times b$ leaves the same remainder as $r_1 \times r_2$.

2.3 Set of Residues Z_n and Z_n^* , inverses, tables

★ Asked in BOTH papers: W2025 Q2(b)OR [4M: Z_n, Z_n^* , members of Z_7 and Z_7^*] + S2026 Q2(b)OR [4M: residue set Z_n]

- **Z_n (set of residues):** all possible remainders when dividing by n . $Z_n = \{0, 1, 2, \dots, n-1\}$.
 $Z_7 = \{0, 1, 2, 3, 4, 5, 6\}$.
- **Z_n^* (set of multiplicative inverses):** only those members of Z_n which have a multiplicative inverse, i.e. numbers with $\gcd(a, n) = 1$ (co-prime to n). Zero is never in Z_n^* .
- $Z_7^* = \{1, 2, 3, 4, 5, 6\}$ (7 is prime, so every non-zero number is co-prime to 7).
- Another example: $Z_{10}^* = \{1, 3, 7, 9\}$ (2, 4, 5, 6, 8 share a factor with 10).

Inverses

- **Additive inverse** of a : the number b such that $a + b \equiv 0 \pmod{n}$. Every member of Z_n has one.
- **Multiplicative inverse** of a : the number b such that $a \times b \equiv 1 \pmod{n}$. Exists only if $\gcd(a, n) = 1$.

Z7 member	0	1	2	3	4	5	6
Additive inverse	0	6	5	4	3	2	1
Multiplicative inverse	-	1	4	5	2	3	6

Check: $3 \times 5 = 15 = 2 \times 7 + 1 \rightarrow 3$ and 5 are multiplicative inverses in Z_7 . Z_7 and Z_7^* differ only by the number 0.

Addition and multiplication tables of Z_5 (syllabus 2.3.3)

Addition table of Z_5

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

Multiplication table of Z_5

×	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

In the addition table, 0 appears in every row/column (so every number has an additive inverse). In the multiplication table, 1 appears in every non-zero row (so every non-zero number has a multiplicative inverse).

2.4.1 Euclidean Algorithm for GCD

★ Euclidean algorithm asked in BOTH papers (7M each): W2025 Q1(c) + S2026 Q2(c) OR

Euclidean algorithm finds the **GCD** (greatest common divisor) of two numbers. Idea: $\text{gcd}(a, b) = \text{gcd}(b, a \bmod b)$; keep going until the remainder is 0.

Algorithm

1. Take $a > b$. Set **$r1 = a$, $r2 = b$** .
2. While $r2 > 0$: **$q = r1 \div r2$** (integer division); **$r = r1 - q \times r2$** ; then set $r1 = r2$ and $r2 = r$.
3. When $r2$ becomes 0, **$\text{gcd}(a, b) = r1$** .

```

r1 = a; r2 = b
while r2 > 0:
    q = r1 // r2
    r = r1 - q * r2
    r1 = r2
    r2 = r
gcd = r1

```

Solved: GCD(1064, 940)

q	r1	r2	r
1	1064	940	124
7	940	124	72
1	124	72	52
1	72	52	20
2	52	20	12
1	20	12	8
1	12	8	4
2	8	4	0

When $r2 = 0$, $r1 = 4$. **GCD(1064, 940) = 4.**

Solved: GCD(2740, 1760)

q	r1	r2	r
1	2740	1760	980
1	1760	980	780
1	980	780	200
3	780	200	180
1	200	180	20
9	180	20	0

When $r2 = 0$, $r1 = 20$. **GCD(2740, 1760) = 20.**

2.4.2 Extended Euclidean Algorithm for Multiplicative Inverse

★ Extended Euclidean algorithm asked in BOTH papers (7M each): W2025 Q1(c)OR + S2026 Q2(c)

Extended Euclidean algorithm finds the GCD and the **multiplicative inverse** of b in Z_n . (The paper's s-columns are the 't' columns of the book.)

Algorithm (to find inverse of b in Z_n)

1. Set $r1 = n$, $r2 = b$, $s1 = 0$, $s2 = 1$.
 2. While $r2 > 0$: $q = r1 \div r2$; $r = r1 - q \times r2$; $s = s1 - q \times s2$; then shift: $r1 = r2$, $r2 = r$, $s1 = s2$, $s2 = s$.
 3. When $r2 = 0$: if $r1 = 1$ the inverse exists and **inverse** = $s1 \bmod n$. If $r1 \neq 1$, no inverse exists.
- ```
r1, r2 = n, b ; s1, s2 = 0, 1
while r2 > 0:
 q = r1 // r2
 r, s = r1 - q*r2, s1 - q*s2
 r1, r2, s1, s2 = r2, r, s2, s
if r1 == 1: inverse = s1 % n
```

### Solved: multiplicative inverse of 19 in $Z_{35}$ ( $n = 35$ , $b = 19$ )

| q | r1 | r2 | r  | s1 | s2  | s   |
|---|----|----|----|----|-----|-----|
| 1 | 35 | 19 | 16 | 0  | 1   | -1  |
| 1 | 19 | 16 | 3  | 1  | -1  | 2   |
| 5 | 16 | 3  | 1  | -1 | 2   | -11 |
| 3 | 3  | 1  | 0  | 2  | -11 | 35  |

$r1 = 1 = 1$ , so the inverse exists.  $s1 = -11$ , so inverse =  $-11 \bmod 35 = 24$ .

**Check:**  $19 \times 24 = 456 = 13 \times 35 + 1 \rightarrow 456 \bmod 35 = 1 \checkmark$

### Solved: multiplicative inverse of 23 in $Z_{100}$ ( $n = 100$ , $b = 23$ )

| q | r1  | r2 | r | s1 | s2  | s   |
|---|-----|----|---|----|-----|-----|
| 4 | 100 | 23 | 8 | 0  | 1   | -4  |
| 2 | 23  | 8  | 7 | 1  | -4  | 9   |
| 1 | 8   | 7  | 1 | -4 | 9   | -13 |
| 7 | 7   | 1  | 0 | 9  | -13 | 100 |

$r1 = 1 = 1$ , so the inverse exists.  $s1 = -13$ , so inverse =  $-13 \bmod 100 = 87$ .

**Check:**  $23 \times 87 = 2001 = 20 \times 100 + 1 \rightarrow 2001 \bmod 100 = 1 \checkmark$

## Last-minute revision sheet - Unit 2

- Division algorithm:  $a = q \times n + r$ ,  $0 \leq r < n$ .
- $(a \pm b) \bmod n = [(a \bmod n) \pm (b \bmod n)] \bmod n$ ;  $(a \times b) \bmod n = [(a \bmod n) \times (b \bmod n)] \bmod n$ .
- $Z_n = \{0, \dots, n-1\}$ ;  $Z_n^*$  = members with  $\gcd(a, n) = 1$ .  $Z_7^* = \{1, \dots, 6\}$ .
- Euclid table columns:  $q$ ,  $r_1$ ,  $r_2$ ,  $r$ . Start  $r_1 = a$ ,  $r_2 = b$ . GCD = last non-zero  $r_1$ .
- Ext. Euclid:  $r_1 = n$ ,  $r_2 = b$ ,  $s_1 = 0$ ,  $s_2 = 1$ ;  $s = s_1 - q \times s_2$ . If  $r_1 = 1$ , inverse =  $s_1 \bmod n$ .